Electrons

01 What is an electron?



An atom has a nucleus with electrons around it.

Look closer. Find the small electron beads around the model's center.

A little explanation. An electron is a tiny bit of matter carrying negative electric charge. It has far less mass than a proton or neutron. An atom has a central nucleus with protons and usually neutrons, and electrons moving around it. Their motion is not a set of perfect circles.

Try this. Point to the nucleus, then the electrons around it.

02 What changes when we rub a balloon?



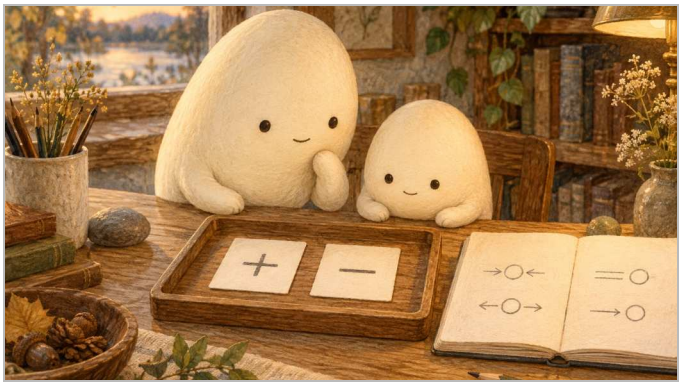
Rubbing some materials together can transfer electrons.

Look closer. A few light paper bits lift toward the balloon.

A little explanation. Rubbing a balloon on wool can transfer some electrons to the balloon. It gains negative charge; the wool loses electrons and becomes positive. Nearby paper has charges that shift a little, so the closer side is pulled toward the balloon.

Try this. With Big leading, rub a balloon on a sweater, then hold it near paper bits on a tray. Keep fingers out of outlets, and keep hands dry around plugs and switches. Big leads every demo.

03 When do charges pull or push?



Opposite charges pull together; like charges push apart.

Look closer. The cards show a plus for positive and a minus for negative.

A little explanation. Opposite charges pull together. Two positive charges push apart, and two negative charges push apart. Losing some negative electrons can leave a material with more positive charge than negative charge.

Try this. Use a plus and a minus card to explain a pull; use two minus cards to explain a push.

04 Do electrons travel in a lamp's wire too?



In a closed circuit, electrons move through the wire.

Look closer. The small lamp has a complete wire loop back to the battery.

A little explanation. Yes. Some electrons can move through metal wires. A battery gives energy to a closed circuit so electric current flows and the lamp lights. Electrons are already in the wires; the lamp does not eat them. Static cling and current involve the same tiny bits moving in different ways.

Try this. Trace the kit's complete loop from the battery through the lamp and back.